

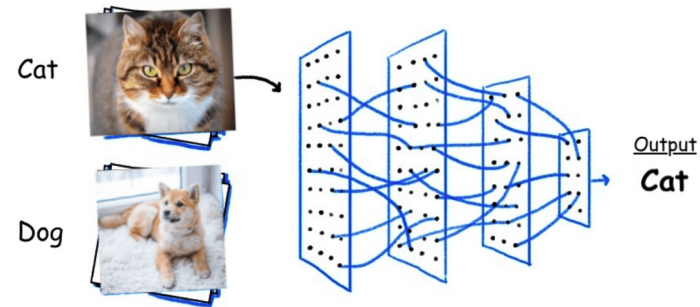


Module 2

Binary Classification



Binary Classification



Types of Classification Problems

- The 3 main classification problems are:

Binary Classification



- Spam
- Not spam

Multiclass Classification



- Dog
- Cat
- Horse
- Fish
- Bird
- ...

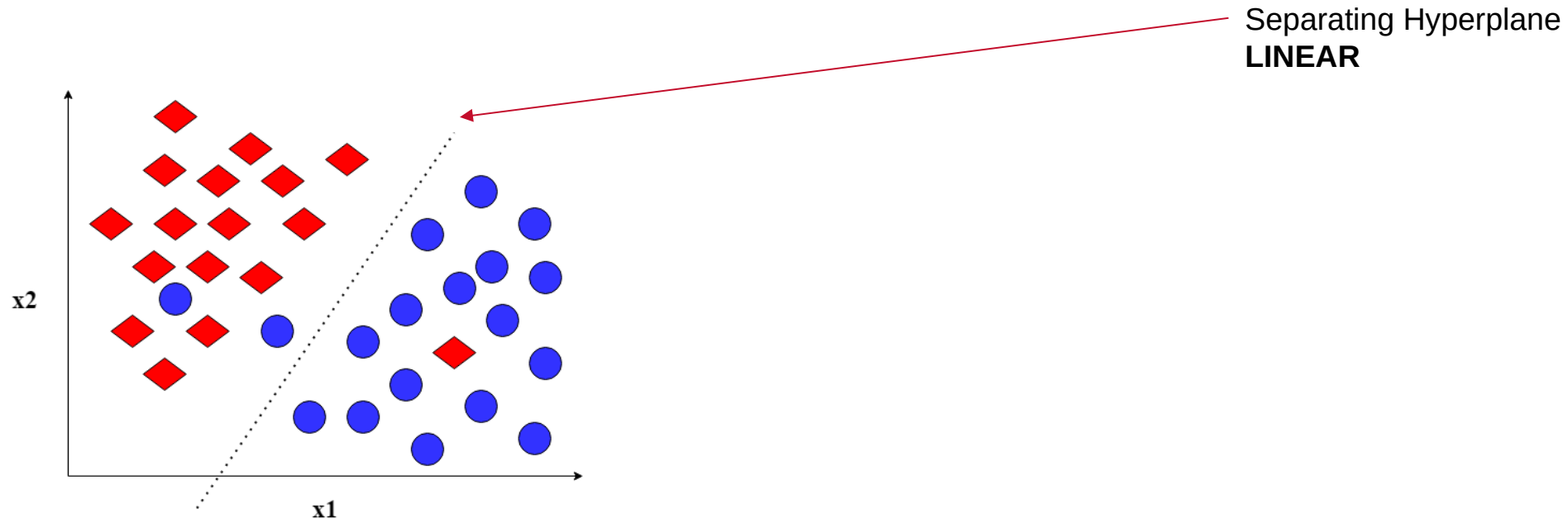
Multi-label Classification



- Dog
- Cat
- Horse
- Fish
- Bird
- ...

Binary Classification

- is the task of classifying the elements of a set into one of two groups (each called class).



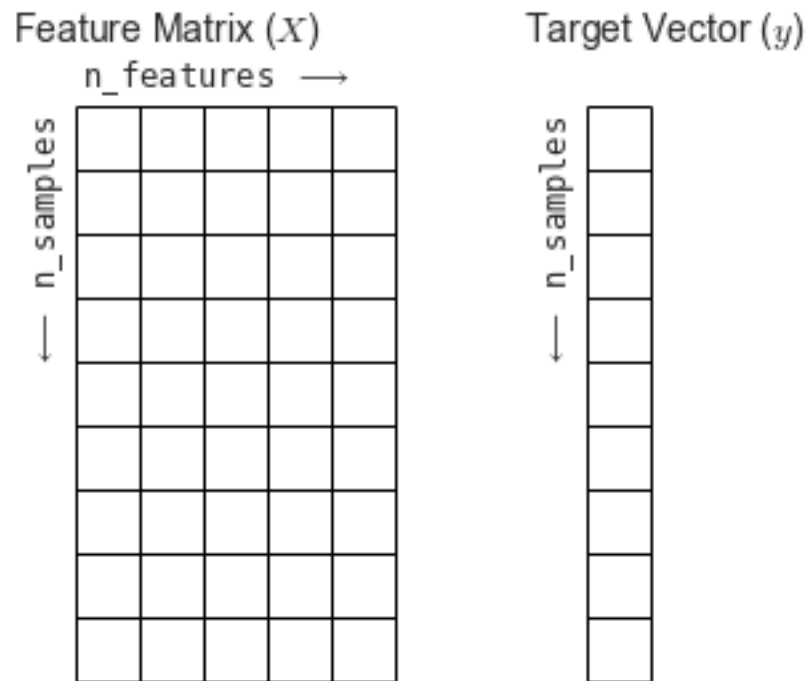
Binary Classification

- is the task of classifying the elements of a set into one of two groups (each called class).



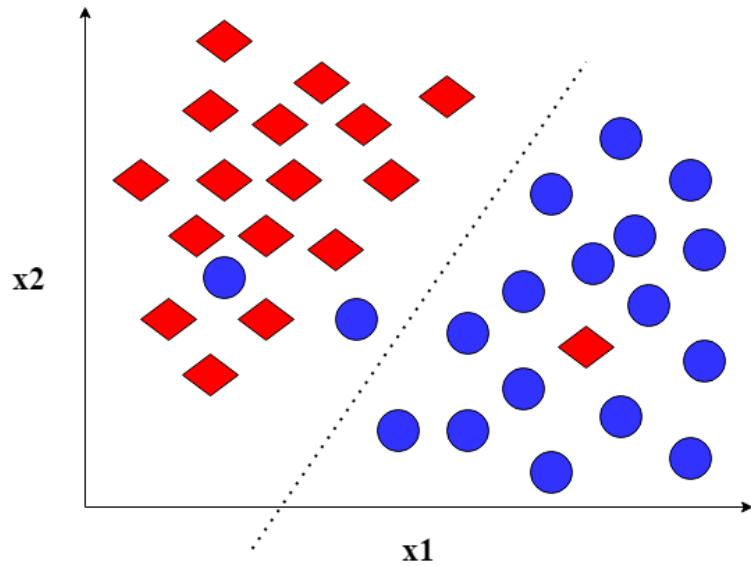
Binary Classification

- is the task of classifying the elements of a set into one of two groups (each called class).



Binary Classification - Propensities

- Linear and non-linear predictors output the probability of belonging to a particular class ($\hat{p}$)



Binary Cross-Entropy

- The most common **loss** function in binary classification

$$-(y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$
1	0.775
0	0.116
0	0.039
0	0.070

How do we determine the **distance** between this two vectors?

Binary Cross-Entropy

- The most common **loss** function in binary classification

$$-(y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$
1	0.775
0	0.116
0	0.039
0	0.070

$$-(1 \log 0.775 + (1 - 1) \log(1 - 0.775)) = 0.2549$$

Binary Cross-Entropy

- The most common **loss** function in binary classification

$$-(y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$
1	0.775
0	0.116
0	0.039
0	0.070

$$-(0 \log 0.116 + (1 - 0) \log(1 - 0.116)) = 0.1233$$

Binary Cross-Entropy

- The most common **loss** function in binary classification

$$-(y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$
1	0.775
0	0.116
0	0.039
0	0.070

$$-(0 \log 0.039 + (1 - 0) \log(1 - 0.039)) = 0.0398$$

Binary Cross-Entropy

- The most common **loss** function in binary classification

$$-(y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$
1	0.775
0	0.116
0	0.039
0	0.070

$$-(0 \log 0.070 + (1 - 0) \log(1 - 0.070)) = 0.0726$$

Binary Cross-Entropy

- The binary cross-entropy cost function is the average of the losses

$$J(\mathbf{w}) = -\frac{1}{n} \sum (y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

y	$\hat{p}$	
1	0.775	0.2549 +
0	0.116	0.1233 +
0	0.039	0.0398 +
0	0.070	0.0726 = 0.4906/4 = 0.12265

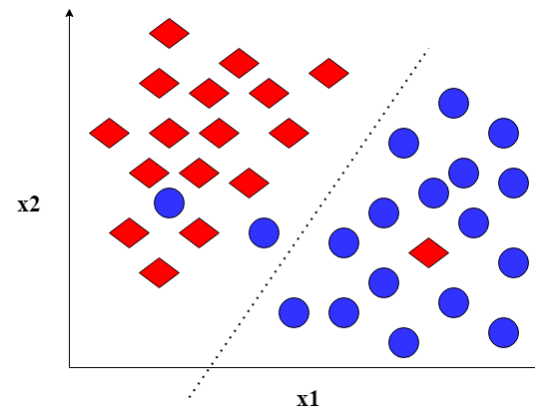
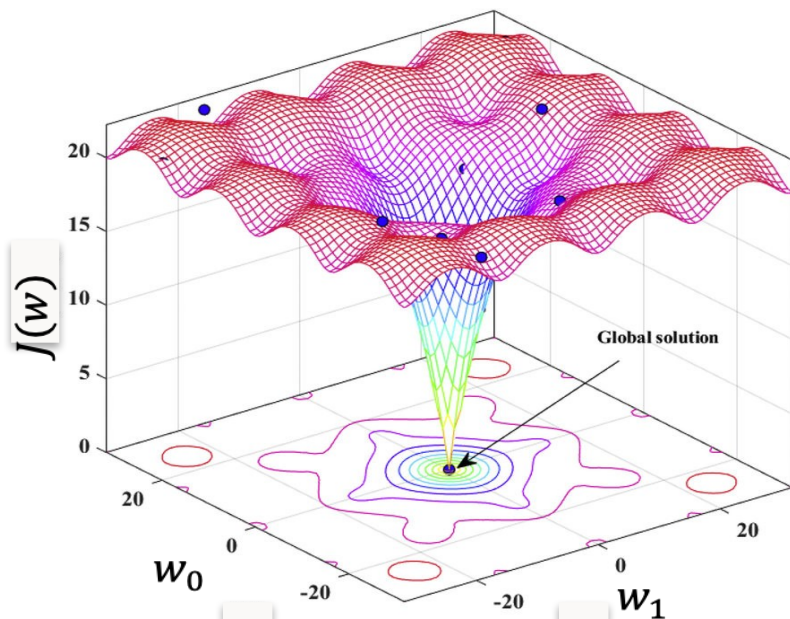
Binary Cross-Entropy

- The binary cross-entropy cost function is the average of the losses

$$J(w) = -\frac{1}{n} \sum (y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

Example: **LOGISTIC REGRESSION**

$$\hat{p} = \frac{1}{1 + e^{-(w_0 + w_1 x_1 + \dots)}} = \frac{1}{1 + e^{Xw}}$$





Python

Metrics

- ❑ Used to determine if model is performing well

Scoring	Function	Comment
Classification		
'accuracy'	<code>metrics.accuracy_score</code>	
'balanced_accuracy'	<code>metrics.balanced_accuracy_score</code>	
'top_k_accuracy'	<code>metrics.top_k_accuracy_score</code>	
'average_precision'	<code>metrics.average_precision_score</code>	
'neg_brier_score'	<code>metrics.brier_score_loss</code>	
'f1'	<code>metrics.f1_score</code>	for binary targets
'f1_micro'	<code>metrics.f1_score</code>	micro-averaged
'f1_macro'	<code>metrics.f1_score</code>	macro-averaged
'f1_weighted'	<code>metrics.f1_score</code>	weighted average
'f1_samples'	<code>metrics.f1_score</code>	by multilabel sample
'neg_log_loss'	<code>metrics.log_loss</code>	requires <code>predict_proba</code> support
'precision' etc.	<code>metrics.precision_score</code>	suffixes apply as with 'f1'
'recall' etc.	<code>metrics.recall_score</code>	suffixes apply as with 'f1'
'jaccard' etc.	<code>metrics.jaccard_score</code>	suffixes apply as with 'f1'
'roc_auc'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovr'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovo'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovr_weighted'	<code>metrics.roc_auc_score</code>	
'roc_auc_ovo_weighted'	<code>metrics.roc_auc_score</code>	

Metrics Review: Confusion Matrix

- ❑ The confusion matrix gives a summary of overall performance
- ❑ True Positives, True Negatives, False Positives and False Negatives

n=165	Predicted: NO	Predicted: YES	
	$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10	60
Actual: YES	$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100	105
	55	110	

Metrics Review: Accuracy

- ❑ The accuracy metric measures number of obs predicted correctly
- ❑ It is one of the most important classification metrics

n=165		Predicted: NO	Predicted: YES	
Actual: NO	$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10		60
Actual: YES	$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100		105
	55	110		

$$\text{accuracy} = \frac{n_{1,1} + n_{0,0}}{n}$$

Metrics Review: Recall (Sensitivity)

- ❑ The fraction of all 1's that are classified (predicted) as 1's.

n=165		Predicted: NO	Predicted: YES	
		Actual: NO	Actual: YES	
		$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10	60
		$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100	105
		55	110	

$$\text{sensitivity} = \frac{n_{1,1}}{n_{1,1} + n_{1,0}}$$

Metrics Review: Precision

- ❑ The fraction of predicted values that are correct.

n=165		Predicted: NO	Predicted: YES	
		Actual: NO	Actual: YES	
		$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10	60
		$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100	105
		55	110	

$$\text{precision} = \frac{n_{1,1}}{n_{1,1} + n_{0,1}}$$

F1 Score

- It is meant to capture both Recall and Precision

$$\text{F1 score} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10	60
Actual: YES	$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100	105
	55	110	

Balanced Accuracy

- We use the balanced accuracy for imbalanced data

$$\text{balanced accuracy} = \frac{1}{2} \left(\frac{n_{0,0}}{n_{0,0} + n_{0,1}} + \frac{n_{1,1}}{n_{1,1} + n_{1,0}} \right)$$

n=165		Predicted: NO	Predicted: YES	
		Actual: NO	Actual: YES	
		$n_{0,0}$ TN = 50	$n_{0,1}$ FP = 10	60
		$n_{1,0}$ FN = 5	$n_{1,1}$ TP = 100	105
		55	110	



Python

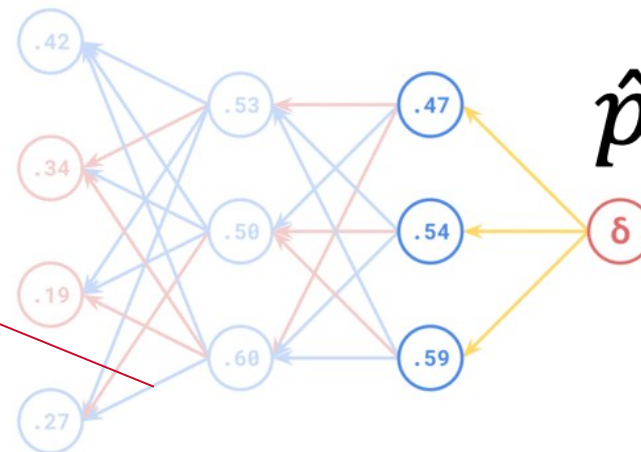
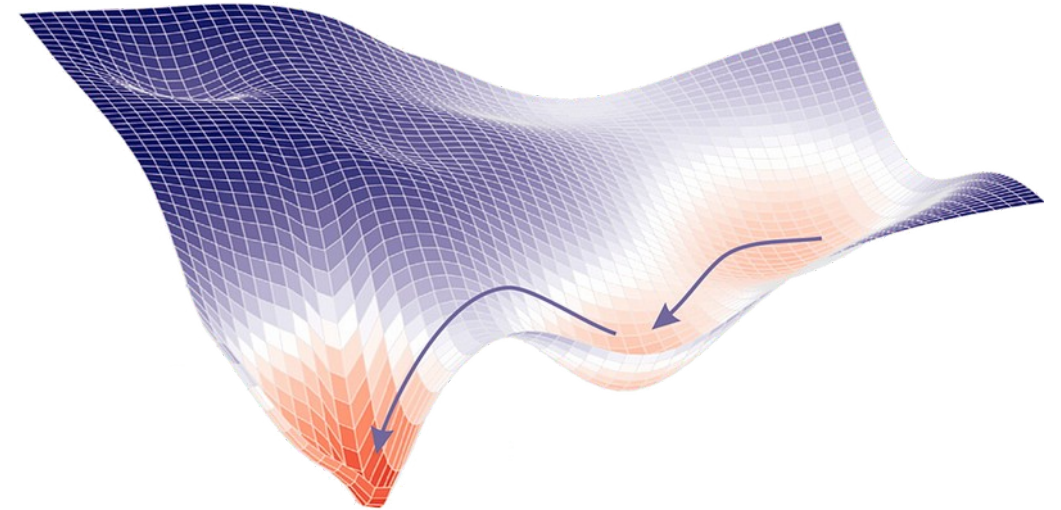
Common Cost Functions Neural Nets

- Binary Cross-Entropy
- Loss Function

$$J(w) = -\frac{1}{n} \sum (y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

- Gradient

$$\nabla J(w) = \hat{p} - y$$



$$\delta_3 \cdot H_2^T = \Delta W_3$$

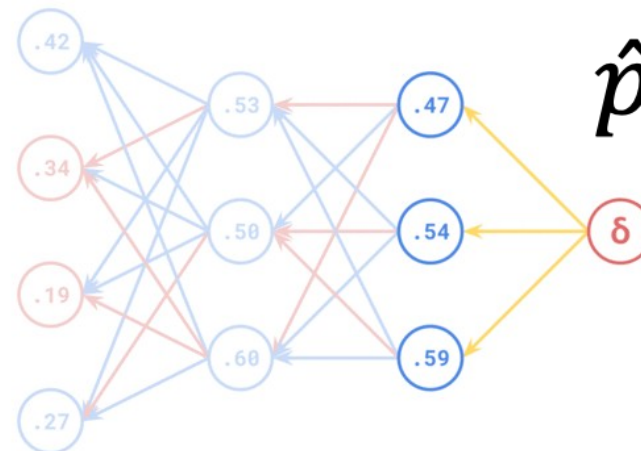
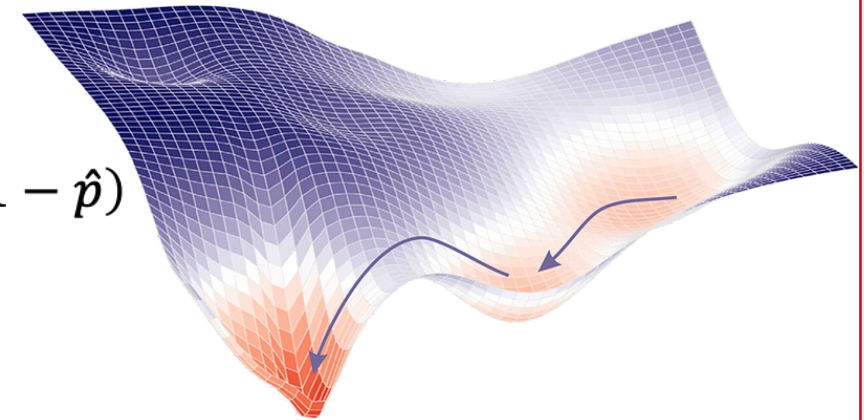
Common Cost Functions Neural Nets

- Binary Focal Cross-Entropy
- Loss Function

$$J(w) = -\frac{1}{n} \sum \alpha(1 - \hat{p})^\gamma y \log(\hat{p}) + (1 - \alpha) \hat{p}^\gamma (1 - y) \log(1 - \hat{p})$$

- Gradient

$$\nabla J(w) = (\gamma(1 - y - \hat{p}) \log(1 - \hat{p}) + (1 - y - \hat{p}) \hat{p}(1 - \hat{p}))$$



$$\delta_3 \cdot H_2^T = \Delta W_3$$

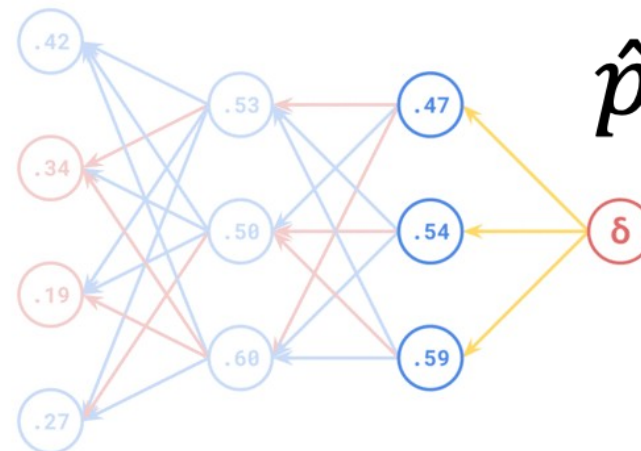
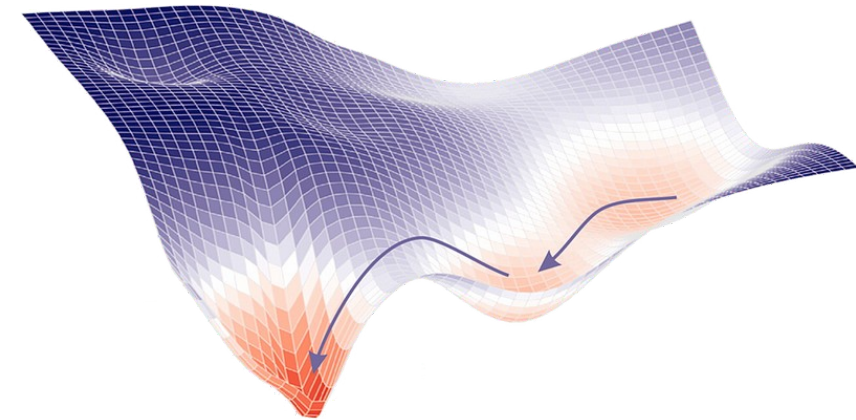
Activation Functions Neural Nets

- Common Activations on Layers

- ☐ ReLu
- ☐ Tanh
- ☐ Sigmoid

- Output Layer Activation

- ☐ Sigmoid
- ☐ Tanh



$$\delta_3 \cdot H_2^T = \Delta W_3$$



Python